

Retail Sales Forecasting & Demand Analysis Report

Executive Summary

This report presents a machine learning-based sales forecasting model built on historical retail transaction data. The objective is to identify patterns in sales behavior and generate predictions to support demand planning and business decision-making.

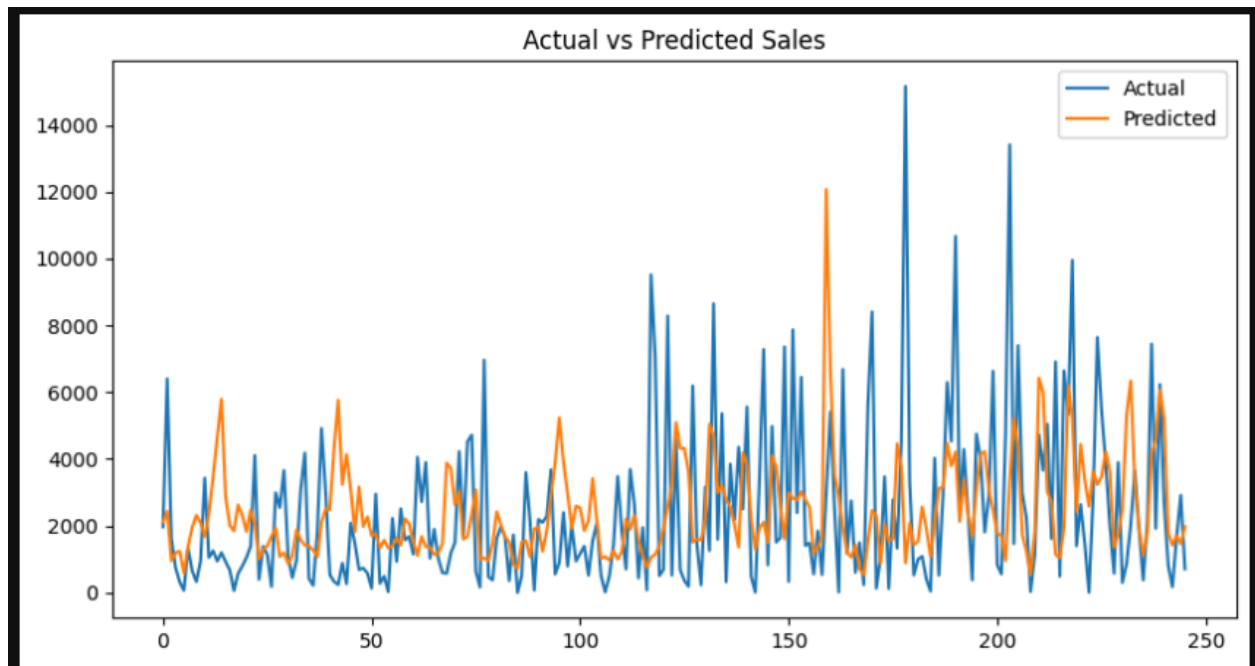
The model leverages time-based features to capture trends and provides insights into key drivers influencing sales performance.

Model Performance

- **Mean Absolute Error (MAE):** 1835.63
- **Root Mean Squared Error (RMSE):** 2576.69

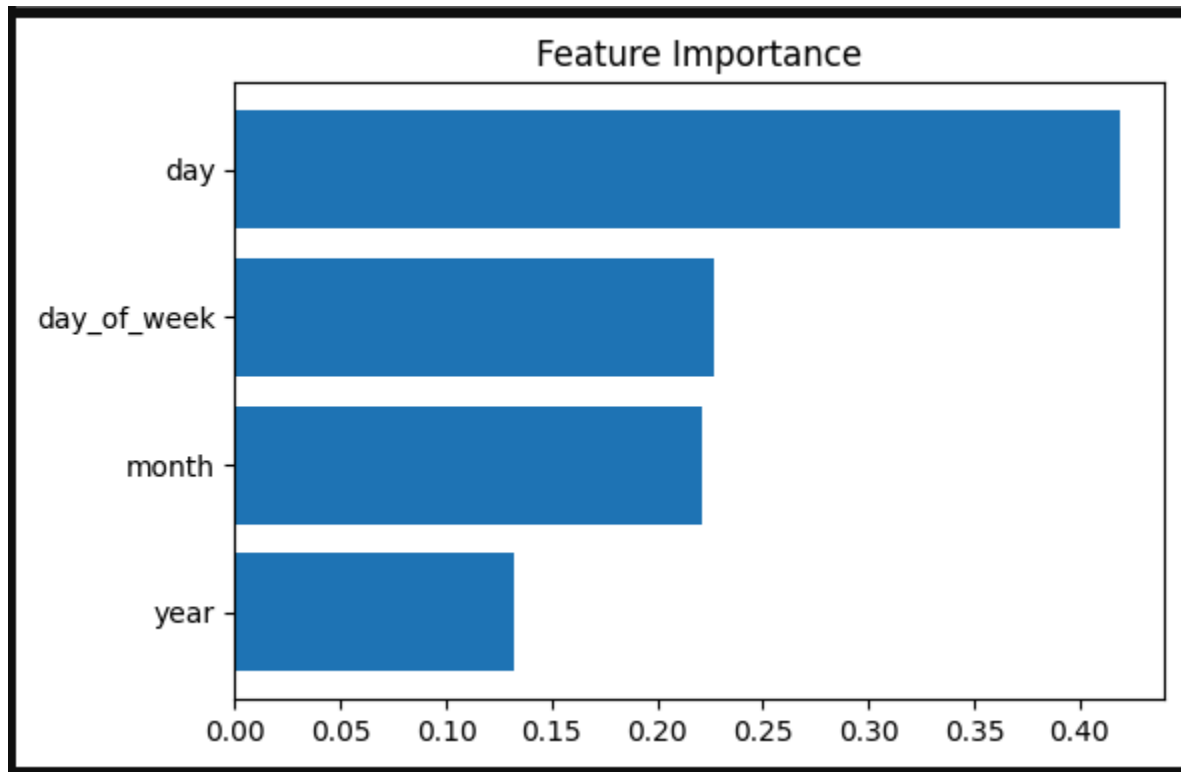
Lower values indicate better prediction accuracy.

Actual vs Predicted Sales Trend



The model captures overall sales trends effectively, though some deviations occur during high-variability periods.

Key Drivers of Sales Prediction



Top Influencing Features:

- Day of the month → 0.419
- Day of the week → 0.227
- Month → 0.221
- Year → 0.132

Sales are primarily driven by calendar-based patterns rather than long-term yearly trends.

Key Insights

- Sales patterns are strongly influenced by calendar-based factors
- Day of the month is the most significant predictor of sales
- Weekly and monthly trends play a major role in demand fluctuations
- The model successfully captures general sales trends
- Some high-variation days are not fully predicted due to missing business factors

Business Recommendations

- Align inventory planning with high-demand periods identified in the model
- Schedule marketing campaigns around peak sales days
- Use forecasting outputs for demand planning and operational decisions
- Improve forecasting accuracy by incorporating additional variables such as:
 - Product category
 - Region
 - Promotions and discounts

Limitations

- The model uses only time-based features
- Business variables such as category, region, and promotions are not included
- Sudden demand spikes cannot be fully captured
- Predictions should be used as directional insights rather than exact values

Conclusion

This project demonstrates the application of machine learning for sales forecasting using historical retail data. The model provides meaningful insights into demand patterns and supports data-driven decision-making.

Future improvements can enhance prediction accuracy by integrating more business-specific features and external factors.